SPACEX LAUNCH ANALYSIS

GITHUB LINK

- https://github.com/tarunpawariya/Data-Science-Capstone-Project

•This repository contains all project files including notebooks and analysis
•Includes data preprocessing, SQL queries, and machine learning models
•Provides complete implementation of SpaceX launch analysis

EXECUTIVE SUMMARY

•This project focuses on analyzing SpaceX launch data to predict the success of rocket landings using data science techniques. SpaceX aims to reduce the cost of space travel by reusing rocket boosters, making landing success a critical factor.
•The dataset was collected from multiple sources including the SpaceX API and web scraping from Wikipedia. The data includes important features such as launch site, payload mass, orbit type, flight number, and landing outcome.
•Data preprocessing and wrangling techniques were applied to clean the dataset, handle missing values, and transform variables into a usable format for analysis.
•Exploratory Data Analysis (EDA) was performed using visualizations such as bar charts, pie charts, and scatter plots to identify patterns and trends in launch success rates.
•SQL queries were used to extract meaningful insights from the dataset, such as total launches, success rates, and performance of different launch sites.
•Machine learning models including Logistic Regression, Decision Tree, and Support Vector Machine (SVM) were applied to predict landing success. The models were evaluated based on accuracy.
•The results show that landing success can be predicted with good accuracy, and factors such as launch site and flight number play an important role in determining the outcome.
•This project demonstrates how data analysis and machine learning can help improve decision-making in space missions.

INTRODUCTION

- In this project, SpaceX launch data is analyzed using data science techniques.
- The dataset includes information such as flight number, launch site, payload mass, orbit type, and landing outcome.
- The main focus of the analysis is to determine whether the first stage of the Falcon 9 rocket will land successfully.
- Data is collected using SpaceX API and processed for analysis.
- Exploratory Data Analysis (EDA) is performed to understand patterns in the dataset.
- SQL queries are used to extract insights from the data.
- Machine learning models are applied to predict the landing outcome.

DATA COLLECTION AND WRANGLING

DATA COLLECTION

•Data was collected using the **SpaceX REST API**
•Additional data was obtained through **web scraping from Wikipedia**
•The dataset includes:
•Flight Number
•Launch Site
•Payload Mass
•Orbit Type
•Landing Outcome

DATA WRANGLING

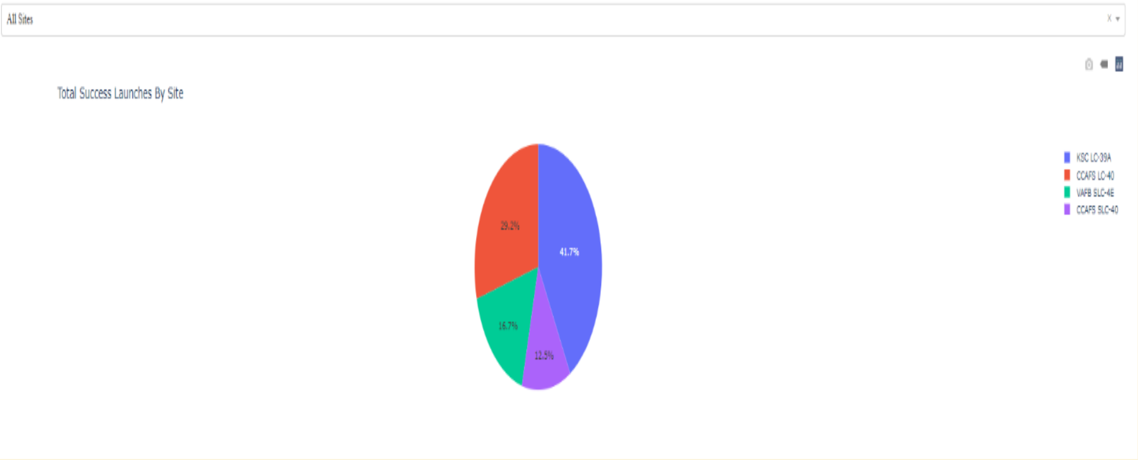
•Removed missing and null values from the dataset
•Converted data into proper formats for analysis
•Filtered relevant columns required for the study
•Created new features such as landing success (1 = success, 0 = failure)
•Cleaned inconsistent and duplicate data

EDA WITH VISUALIZATION

- Exploratory Data Analysis (EDA) was performed to understand the structure and distribution of the SpaceX launch dataset.
- Various visualization techniques were used to analyze the relationship between different features such as flight number, launch site, payload mass, and landing outcome.
- Bar charts were used to compare launch success rates across different launch sites.
- Scatter plots were used to examine the relationship between payload mass and landing success.
- Pie charts were used to visualize the proportion of successful and failed landings.
- The analysis helped in identifying important patterns and trends in the dataset.

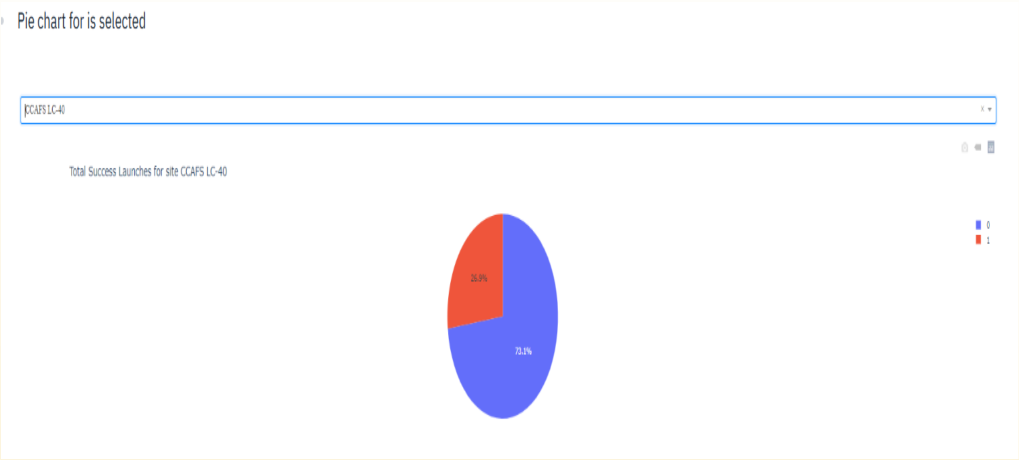
INTERACTIVE DASHBOARD (PLOTLY DASH)

PIE CHART (LAUNCH SITE VS SUCCESS)



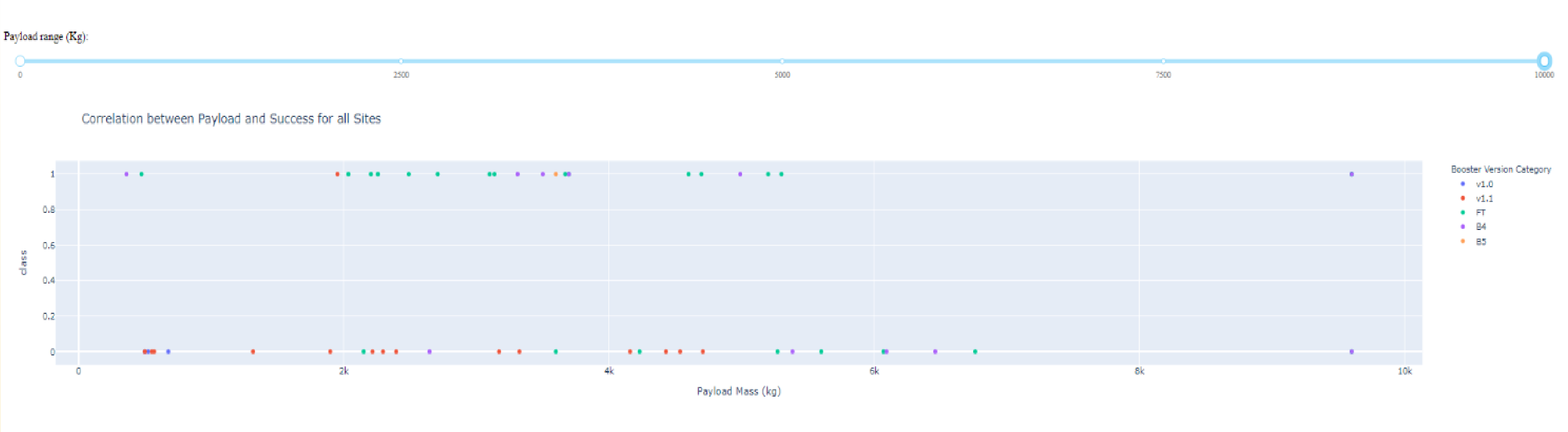
•Displays success vs failure ratio for a selected launch site
•Majority of launches from this site are successful

PIE CHART (SELECTED SITE - CCAFS LC-40)



•Shows distribution of successful launches across different launch sites
•KSC LC-39A has the highest success contribution among all sites

SCATTER PLOT (PAYLOAD VS SUCCESS):



•Shows relationship between payload mass and landing success
•Medium payload range has higher chances of successful landing

PREDICTIVE ANALYSIS

•Machine learning techniques were applied to predict the landing outcome of the SpaceX Falcon 9 first stage.

•The dataset was split into training and testing data for model evaluation.

•The following classification models were used:
•Logistic Regression
•Decision Tree
•Support Vector Machine (SVM)

•Features such as flight number, payload mass, launch site, and orbit type were used as input variables

•The models were trained on historical launch data and used to predict whether the landing would be successful or not.

•Model performance was evaluated using accuracy score.

EDA WITH SQL

- Query 1) SELECT DISTINCT Launch_Site FROM SPACEXTBL;

- Query 2) SELECT COUNT(*)

  FROM SPACEXTBL

  WHERE Landing_Outcome='Success';

- Query 3) SELECT Launch_Site, COUNT(*)

  FROM SPACEXTBL

  GROUP BY Launch_Site;

- Multiple launch sites are used for SpaceX missions

- The number of successful landings is higher compared to failures

- Some launch sites have more launches than others

EXAMPLE OF EDA WITH SQL

Display 5 records where launch sites begin with the string 'CCA'

```
%%sql
SELECT *FROM SPACEXTBL
WHERE "Launch_Site" Like 'CCA%'
LIMIT 5;
```

* sqlite:///my_data1.db
Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

```
%%sql
SELECT DISTINCT "Launch_Site"
FROM SPACEXTBL;
```

* sqlite:///my_data1.db
Done.

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

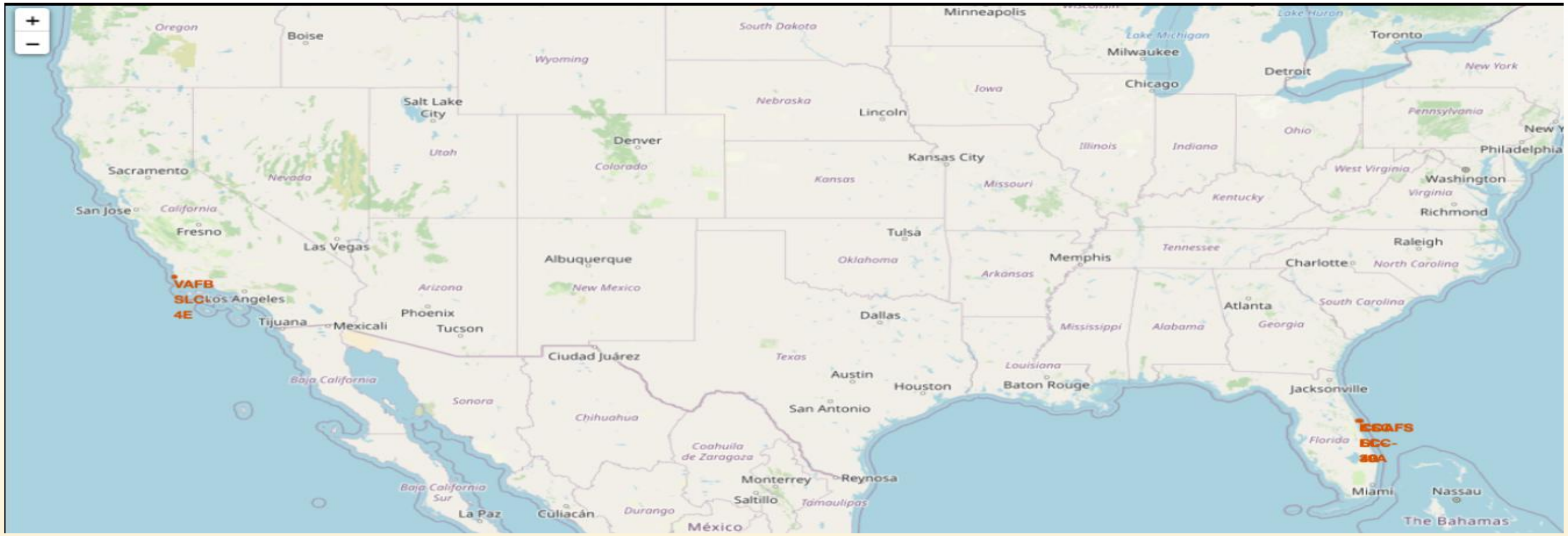
FOLIUM MAP

- import folium

  # Create base map
  spacex_map = folium.Map(location=[29.5, -95], zoom_start=5)

  # Launch sites coordinates
  launch_sites = {
  "KSC LC-39A": [28.573255, -80.646895],
  "CCAFS SLC-40": [28.561857, -80.577366],
  "VAFB SLC-4E": [34.632834, -120.610745]
  }

OUTPUT OF FOLIUM MAP

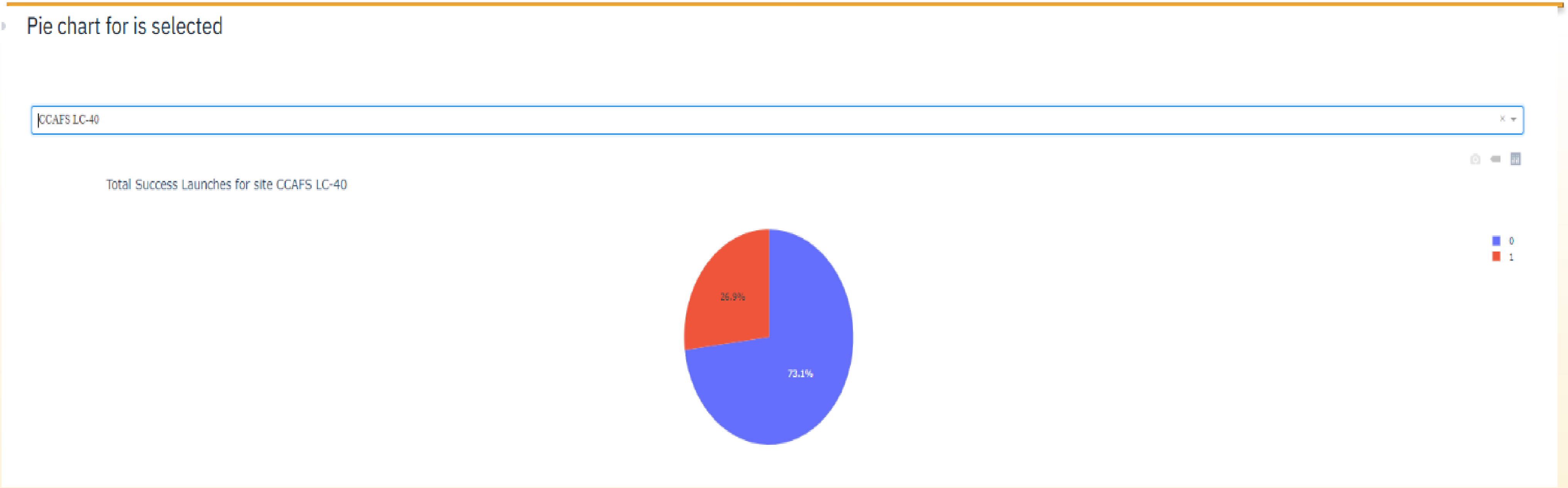


- The map shows the geographical locations of major SpaceX launch sites
- Markers represent KSC LC-39A, CCAFS SLC-40, and VAFB SLC-4E
- These locations are used for different types of space missions

PLOTLY DASH

- An interactive dashboard was developed using Plotly Dash to analyze SpaceX launch data.
- The dashboard allows users to select different launch sites and view their success rates.
- A pie chart is used to show the proportion of successful and failed launches for each SpaceX launch site
- A scatter plot is used to analyze the relationship between payload mass and landing success.
- A payload range slider is provided to filter launches based on payload mass.
- The dashboard helps in understanding how different factors such as launch site and payload affect rocket landing success.

EXAMPLE



•Launch sites like **KSC LC-39A and CCAFS SLC-40** show higher success rates
•Payload mass has an impact on landing success, with certain ranges showing better performance

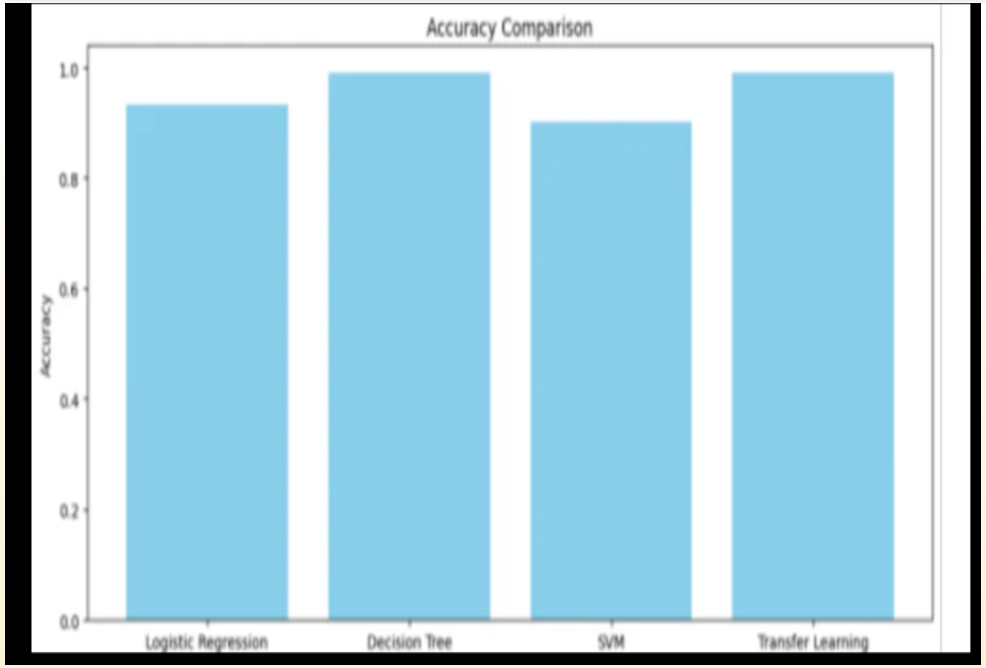
MODEL RESULTS AND EVALUATION

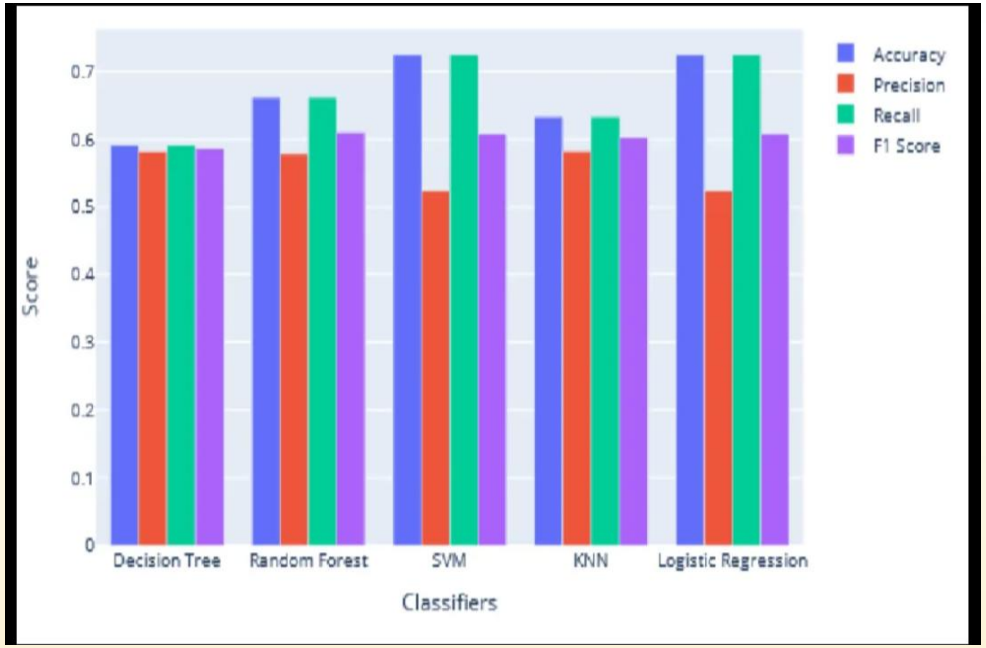
- Different machine learning models were applied to predict the landing success of SpaceX Falcon 9 rockets.
- The models were evaluated using accuracy score to compare their performance.

- The obtained accuracy results are:
- Logistic Regression: **83%**
- Support Vector Machine (SVM): **83%**
- K-Nearest Neighbors (KNN): **84%**
- Decision Tree: **87%**

- Among all the models, the **Decision Tree achieved the highest accuracy (87%)**, making it the best-performing model for this prediction task.
- The results indicate that machine learning models can effectively classify landing outcomes based on features such as launch site, payload mass, and flight number.
- This analysis demonstrates that predictive modeling can help improve decision-making in SpaceX launch operations.

ACCURACY OF MODELS





- Decision Tree performed better because it can capture complex patterns in the dataset more effectively than other models

CONCLUSION

• This project analyzed SpaceX launch data to understand the factors affecting rocket landing success.
• The analysis showed that features such as **launch site, payload mass, and flight number** have a significant impact on landing outcomes.
• Exploratory Data Analysis (EDA) and visualizations helped identify patterns, such as higher success rates at specific launch sites.
• SQL queries provided useful insights into launch performance and distribution across different sites.
• Machine learning models were successfully applied to predict landing outcomes, with **Decision Tree achieving the highest accuracy (87%)**.
• The results confirm that rocket landing success can be predicted with good accuracy using historical data.
• This analysis can support better planning and decision-making for future SpaceX missions.